

INTERNSHIP PROJECT REPORT

Smart Dynamic Sampling Rate (Adaptive Monitoring)

Task 5 – ESP32, FreeRTOS, MQTT and Adafruit IO

A practical adaptive patient-monitoring prototype implemented and tested using ESP32 in the Wokwi simulation environment.

Platform	ESP32 / Wokwi
Monitoring Parameters	Heart Rate, SpO ₂ , Body Temperature
Communication	Wi-Fi and MQTT
Cloud Dashboard	Adafruit IO
Control Modes	AUTO and MANUAL

Prepared from the implemented Task 5 system and supplied test screenshots

1. Objective

The objective of this task is to develop an adaptive patient-monitoring system in which the sampling frequency changes according to monitored patient parameters. The system provides real-time monitoring of Heart Rate (HR), SpO₂ and body temperature, detects abnormal conditions, increases monitoring frequency during abnormal conditions, and reduces the sampling frequency during stable periods. It also provides remote manual sampling-rate control through an Adafruit IO dashboard.

2. System Overview

The prototype uses an ESP32 as the main controller. Three potentiometers in Wokwi simulate patient parameters. An OLED provides local information, while Wi-Fi and MQTT provide communication with Adafruit IO. The system supports two operating modes: AUTO mode, where the sampling interval is selected according to patient condition, and MANUAL mode, where the sampling interval is selected remotely from the dashboard.

3. Simulated Patient Parameters and Thresholds

Parameter	Simulated Range	Normal Condition
Heart Rate	40–140 BPM	60–100 BPM
SpO ₂	85–100 %	≥ 95 %
Body Temperature	34.0–40.0 °C	36.0–37.5 °C

4. Adaptive Sampling Logic

In AUTO mode, the system evaluates the monitored parameters and changes the sampling interval based on the patient condition. A stable condition uses a 30-second interval, while any abnormal condition causes high-frequency monitoring at 5-second intervals.

Operating Mode	Patient Condition	Active Sampling
AUTO	Stable	30 seconds
AUTO	Abnormal	5 seconds
MANUAL	User selected	5–60 seconds

5. Hardware and Software

- ESP32 microcontroller
- Three potentiometers for Wokwi-based simulation of patient parameters
- SSD1306 OLED display
- Wokwi simulation environment
- FreeRTOS task-based monitoring structure
- Wi-Fi connectivity
- MQTT communication
- Adafruit MQTT Library
- Adafruit IO dashboard
- Arduino CLI / VS Code development environment

6. Dashboard Implementation

The Adafruit IO dashboard provides visualization of the three patient parameters and remote sampling control. The dashboard contains Heart Rate, SpO₂ and Temperature gauges, a Patient Status indicator, a sampling interval control and a Sampling Mode control. The dashboard communicates with the ESP32 through MQTT feeds.

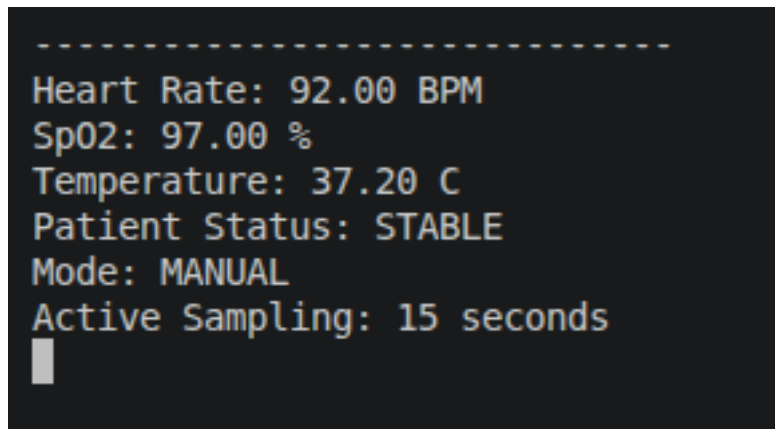


Figure 1: Adafruit IO dashboard showing Heart Rate, SpO₂, Temperature, Patient Status, sampling interval and Sampling Mode.

7. Manual Sampling Mode

The manual mode allows the sampling interval to be selected remotely from the dashboard within the configured 5–60 second range. The supplied Serial Monitor evidence shows a stable patient condition operating in MANUAL mode with an active sampling interval of 15 seconds. Another supplied log demonstrates reception of a dashboard sampling interval command of 25 seconds.

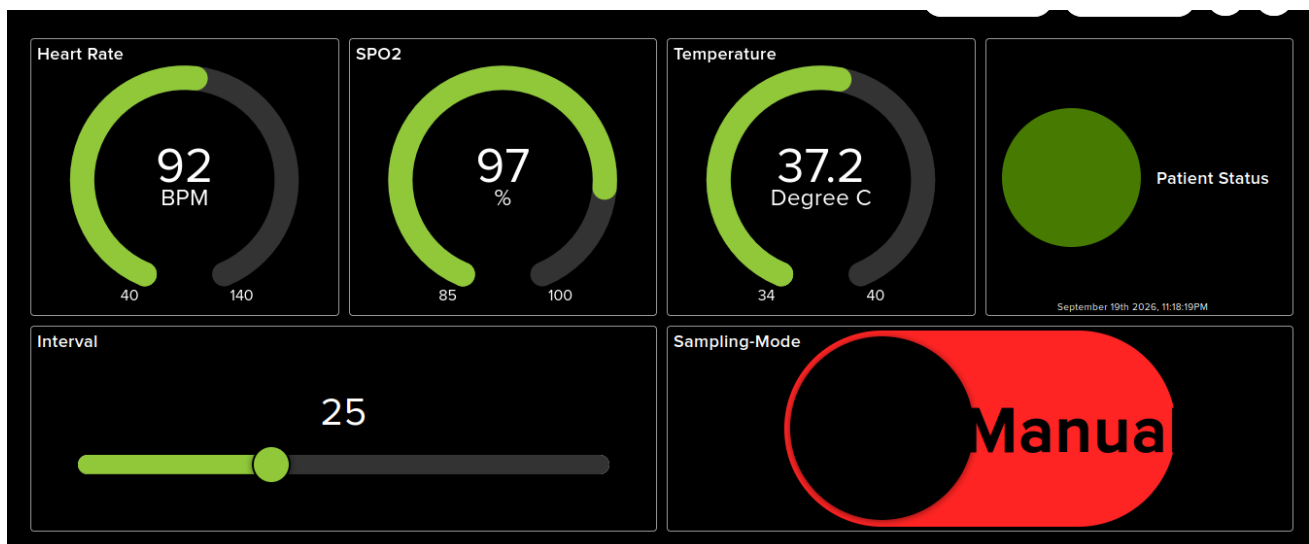


Figure 2: Serial Monitor showing MANUAL mode with a 15-second active sampling interval.

```
-----  
Heart Rate: 92.00 BPM  
SpO2: 97.00 %  
Temperature: 37.20 C  
Patient Status: STABLE  
Mode: AUTO  
Active Sampling: 30 seconds  
Sampling mode: MANUAL  
New manual sampling interval: 25 seconds  
█
```

Figure 3: Adafruit IO dashboard showing the manual interval control and Manual mode.

```
-----  
Heart Rate: 92.00 BPM  
SpO2: 97.00 %  
Temperature: 35.40 C  
Patient Status: ABNORMAL  
Mode: AUTO  
Active Sampling: 5 seconds  
█
```

Figure 4: Serial Monitor showing reception of the MANUAL mode and a new sampling interval command.

8. Abnormal Condition Detection

Testing was performed by deliberately moving the simulated patient parameters outside their normal ranges. When an abnormal condition was detected in AUTO mode, the system reported ABNORMAL status and changed the active sampling interval to 5 seconds.

```
-----  
Heart Rate: 92.00 BPM  
SpO2: 91.00 %  
Temperature: 36.70 C  
Patient Status: ABNORMAL  
Mode: AUTO  
Active Sampling: 5 seconds  
█
```

Figure 5: Abnormal temperature condition resulting in 5-second automatic sampling.

```
-----  
Heart Rate: 59.00 BPM  
SpO2: 96.00 %  
Temperature: 36.70 C  
Patient Status: ABNORMAL  
Mode: AUTO  
Active Sampling: 5 seconds  
█
```

Figure 6: Abnormal condition with Heart Rate below the normal lower threshold.

```
-----  
Heart Rate: 87.00 BPM  
SpO2: 96.00 %  
Temperature: 36.70 C  
Patient Status: STABLE  
Mode: AUTO  
Active Sampling: 30 seconds  
█
```

Figure 7: Stable condition restored after the monitored parameters returned to the normal range.

9. Testing and Validation

Test Case	Condition	Expected / Verified Result	Result
TC-01	Normal HR, SpO ₂ and temperature	STABLE; 30 s in AUTO	PASS
TC-02	SpO ₂ < 95%	ABNORMAL; 5 s	PASS
TC-03	HR > 100 BPM	ABNORMAL; 5 s	PASS
TC-04	HR < 60 BPM	ABNORMAL; 5 s	PASS

Test Case	Condition	Expected / Verified Result	Result
TC-05	Temperature > 37.5 °C	ABNORMAL; 5 s	PASS
TC-06	Temperature < 36.0 °C	ABNORMAL; 5 s	PASS
TC-07	MANUAL at 5 s	Active sampling = 5 s	PASS
TC-08	MANUAL at 30 s	Active sampling = 30 s	PASS
TC-09	MANUAL at 60 s	Active sampling = 60 s	PASS
TC-10	MANUAL → AUTO transition	Automatic adaptive control restored	PASS

10. Observed Test Evidence

The supplied screenshots provide evidence of normal operation, manual sampling control, abnormal-condition detection, automatic 5-second sampling, and restoration to 30-second sampling after the patient parameters return to normal. The Serial Monitor and Adafruit IO dashboard together demonstrate the local and remote sides of the system.

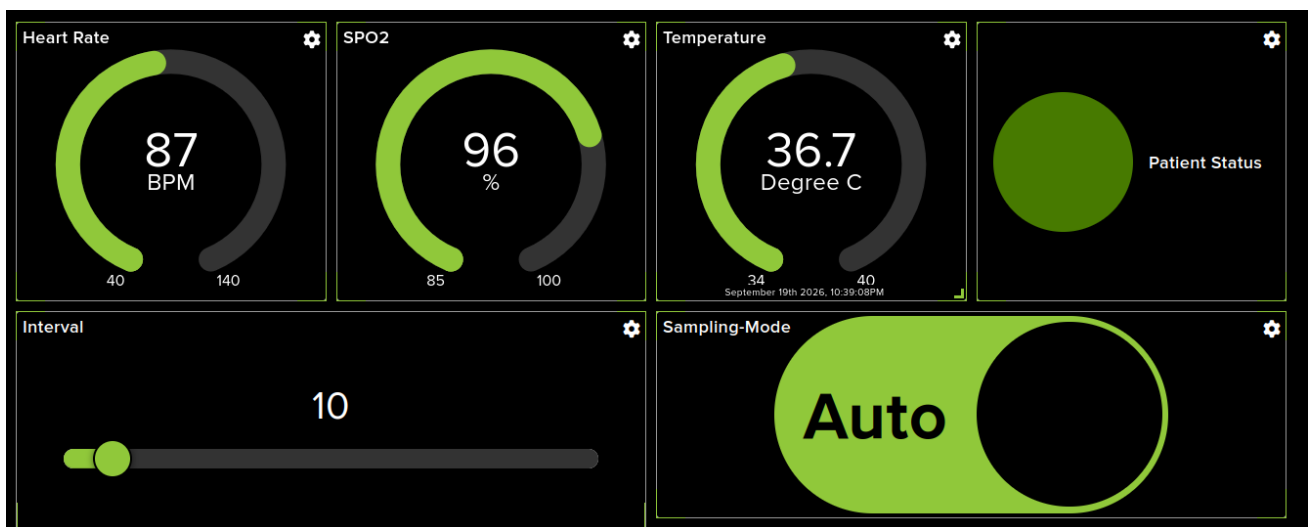


Figure 8: Additional dashboard view used during functional testing.

11. Result

The Smart Dynamic Sampling Rate (Adaptive Monitoring) system was successfully implemented and functionally tested using an ESP32-based Wokwi simulation. The system demonstrated real-time monitoring of Heart Rate, SpO₂ and body temperature; abnormal-condition detection; automatic transition between 30-second and 5-second sampling in AUTO mode; manual sampling control from 5 to 60 seconds; AUTO/MANUAL operation; OLED display; and MQTT-based communication with an Adafruit IO dashboard.

12. Conclusion

The completed prototype demonstrates an adaptive monitoring approach in which monitoring frequency is increased during abnormal conditions and reduced during stable operation. The combination of FreeRTOS task-based processing, ESP32, MQTT and Adafruit IO provides a practical foundation for a connected patient-monitoring prototype. Functional tests covering normal conditions, abnormal Heart Rate, abnormal SpO₂, abnormal temperature, manual sampling intervals and mode transitions were completed successfully.

Note: The report documents the implemented simulation and observed functional behavior. The patient parameters are simulated using Wokwi potentiometers and are not medical-grade sensor measurements.